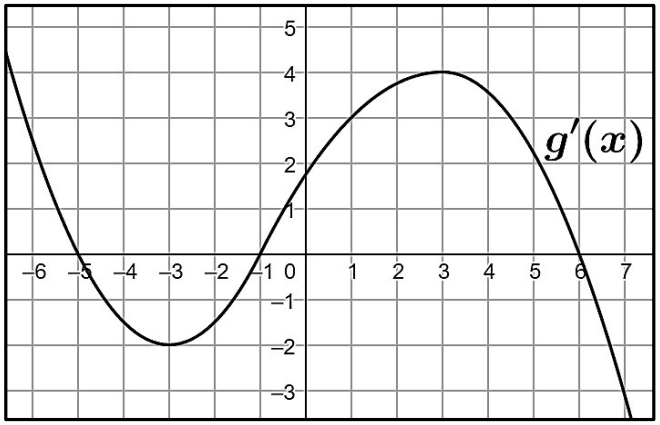
Worksheet 5-6 B Concavity and Points of Inflection #2
AP Calculus Name_____




1. The graph of $g'(x)$, the derivative of $g$, is continuous and differentiable for all real numbers and is given above. $g'(x)$ has horizontal tangents at $x = -3$ and $x = 3$.

(A) On what interval(s) is $g(x)$ decreasing? Give a reason for your answer.

(B) At which value(s) of $x$ does $g(x)$ have a relative maximum? Explain your reasoning.

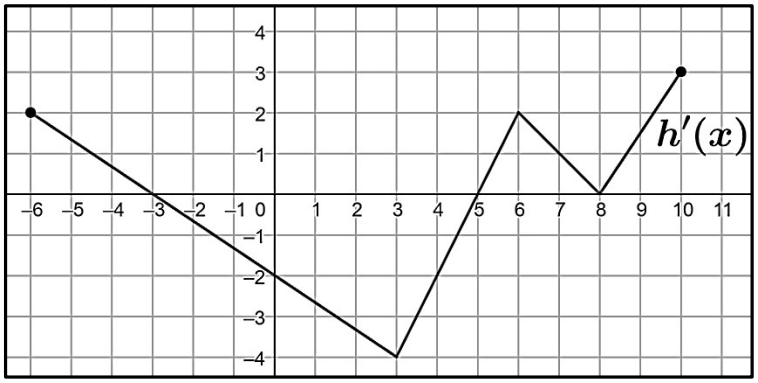
(C) On what interval(s) is $g(x)$ concave up? Give a reason for your answer.

(D) On what interval(s) is $g(x)$ concave down? Explain your reasoning.

(E) At which value(s) of $x$ does $g(x)$ have a point of inflection? Explain your reasoning.

(F) On what interval(s) is $g(x)$ both increasing and concave down? Give a reason for your answer.

(G) On what interval(s) is $g(x)$ both decreasing and concave up? Give a reason for your answer.




2. The graph of $h'(x)$, the derivative of $h$, consists of four line segments and is given above.

(A) On what interval(s) is $h(x)$ increasing? Give a reason for your answer.

(B) On what interval(s) is $h(x)$ decreasing? Give a reason for your answer.

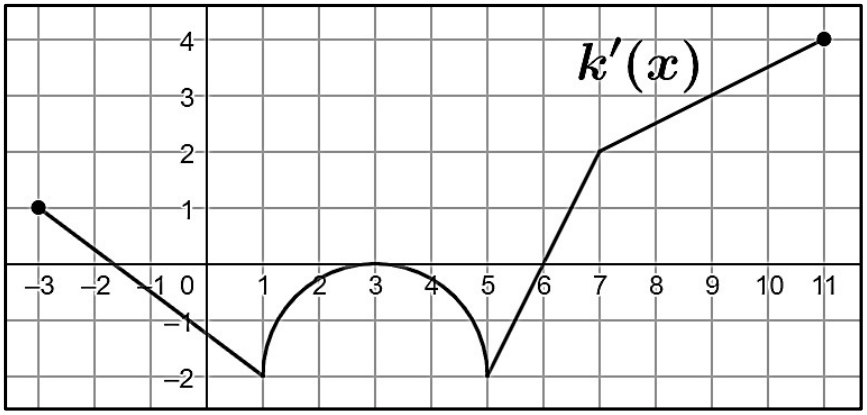
(C) At which value(s) of $x$ does $h(x)$ have a relative minimum? Explain your reasoning.

(D) At which value(s) of $x$ does $h(x)$ have a relative maximum? Explain your reasoning.

(E) On what interval(s) is $h(x)$ concave up? Give a reason for your answer.

(F) On what interval(s) is $h(x)$ concave down? Explain your reasoning.

(G) At which value(s) of $x$ does $h(x)$ have a point of inflection? Explain your reasoning.




3. The graph of $k'(x)$, shown above, consists of three line segments and a semi circle.

(A) At which value(s) of $x$ does $k(x)$ have a point of inflection? Explain your reasoning.

(B) Order the following values from least to greatest: $k''(3)$, $k''(4)$, $k''(6)$.

(C) Order the following values from least to greatest: $k''(-2)$, $k''(6)$, $k''(9)$.

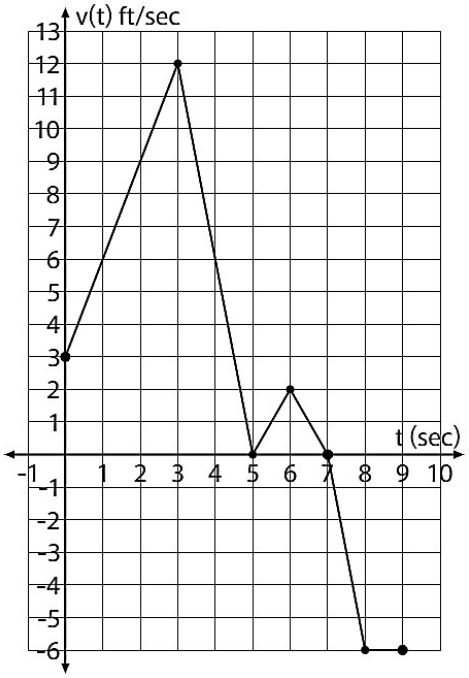
(D) Order the following values from least to greatest: $k'(-2)$, $k'(-1)$, $k'(3)$.

(E) Order the following values from least to greatest: $k'(0)$, $k''(2)$, $k'(6)$.

(F) For each of $k'(5)$ and $k''(5)$, find the value or explain why it does not exist.

(G) For each of $k'(0)$ and $k''(0)$, find the value or explain why it does not exist.

4. The graph above represents the velocity $v$, in feet per second, of a particle moving along the $x$-axis over the time interval for $0 \le t \le 9$ seconds.




(A) At $t = 4$ seconds, is the particle moving to the right or left? Justify.

(B) At what time(s) is the particle at rest? Justify.

(C) At what time(s) does the particle change direction? Justify.

(D) On what open intervals $0 < t < 9$ is the particle moving left? Justify.

(E) What is the acceleration of the particle at $t = 4$ seconds? Show the work that leads to your answer.

(F) On what open intervals $0 < t < 9$ is the acceleration of the particle positive? Justify.

(G) What is the **average acceleration** of the particle over the interval [3,6] seconds? Show the computations that lead to your answer and indicate units of measure.

(H) On what open intervals $0 < t < 9$ is the **speed** of the particle decreasing? Justify.